

Semantic Binding over Syntactic Binding

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Outline

- 1 Introduction
- 2 Problems with syntactic binding
 - Pseudoclefts
 - Wh-questions
- 3 Contexts, abstraction over contexts, and semantic binding
 - Scopes of operators and semantic binding
 - Necessity of non-contextual meaning
- 4 Explanation of connectivity
 - Pseudoclefts
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- 5 Conclusion

Syntactic Binding

- Syntactic binding is defined as a structural configuration.
 α binds β iff α c-command β and α & β are coindexed.
- Binding Conditions (Haegemam & Guéron 1999, p. 376)
 - Anaphors (reflexives, reciprocals) must be locally bound. (BC A)
 - Pronouns must be locally free. (BC B)
 - R-expressions must be free everywhere. (BC C)
- Local domain of a DP: the minimal clause (TP) dominating the DP.

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Connectivity Effects (I)

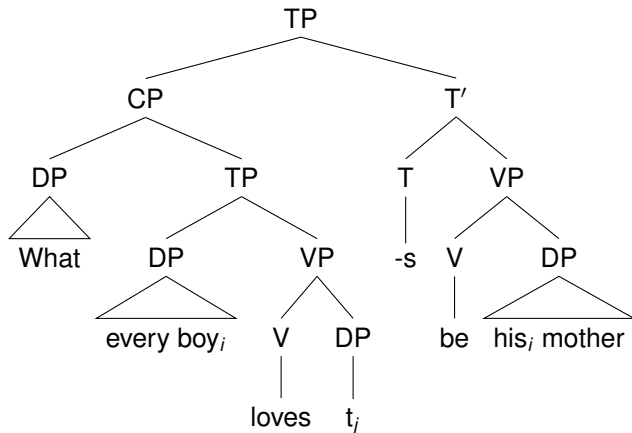
- Pseudoclefts show Connectivity Effects of variable binding and BCs.

Examples 1. (Pseudoclefts)

a. What_{*j*} [every boy]_{*i*} loves t_{*j*} is his_{*i*} mother.

b. What Bill_{*i*} is --- is a nuisance to himself_{*i,*k*}/him_{**i,j*}.

Connectivity Effects (II)



Question in Disguise Theory

- Ross (1972), den Dikken, Meinungen & Wilder (2000), Schlenker (2003): Question in Disguise Theory

Example 2. (Pseudocleft)

What I did then was call the grocer.

= What I did then was $\dagger$ called the grocer. (Ross 1972, (39b))

Example 3. (Pseudocleft)

What [every boy] loves is [every boy] loves his mother.

Example 4. (Pseudocleft)

그가 사랑한 것은 자기의 아버지이다.

*그가 사랑한 것은 그가 자기의 아버지를 사랑하이다.

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Reconstruction required (I)

- Heycock (1995), Fox & Nissenbaum (2004), Spotiche (2017) observed that in wh-questions, bindings facts require reconstruction.

Examples 5. (Wh-questions)

- a. *[Which of each other_{*j*}'s friends]_{*i*} did she remind *t_i* that they_{*j*} saw Bill?
- b. [Which of each other_{*j*}'s friends]_{*i*} did they_{*j*} remind *t_i* that she saw Bill?
- c. [Which of each other_{*j*}'s friends]_{*i*} did she remind Bill that they_{*j*} saw *t_i* ?

Reconstruction required (II)

Example 6. (Wh-questions)

*[Whose examination of John_{*j*}]_{*i*} did he_{*j*} fear *t*_{*i*} ?

⇒ *did he_{*j*} fear [whose examination of John_{*j*}]_{*i*}?

Intervention Effect (I)

- On the other hand, Beck's (2006) Intervention Effect says that a preposed wh-phrase must not be reconstructed.

Examples 7. (Intervention Effect)

- Which museum did [only John_F] visit?
- ??When did [only John_F] visit which museum?

Examples 8. (Intervention Effect)

- 누구를_i 철수만 t_i 만났냐?
- ??철수만 누구를 만났냐?

Intervention Effect (II)

- This means the binding facts in Examples 7 and 8 must be explained without reconstruction.
Sportiche (2017) says, “Reconstruction effects can be viewed as special cases of the more general category of Connectivity effects.”
- Both pseudoclefts and wh-questions involve connectivity effects.
- Connectivity effects are not explained in syntax.

Basic Idea of Proposal (I)

Examples 1a. (Pseudoclefts)

$\llbracket \text{What}_i [\text{every boy}]_i \text{ loves } t_i \text{ is his}_i \text{ mother} \rrbracket^{wg}$

= The **X** such that $\forall x [\text{boy}^w(x) \rightarrow \text{love}^w(x, \mathbf{X}^{wg[i/x]})]$ is

$\lambda w' \lambda g' [\llbracket \text{his}_i \text{ mother} \rrbracket^{w'g'}]$

Let's say $\lambda w' \lambda g' [\llbracket \text{his}_i \text{ mother} \rrbracket^{w'g'}]$ is **HM**. Then

= The **X** such that $\forall x [\text{boy}^w(x) \rightarrow \text{love}^w(x, \mathbf{X}^{wg[i/x]})]$ is **HM**

= $\forall x [\text{boy}(x) \rightarrow \text{love}(x, \mathbf{HM}^{wg[i/x]})]$

= $\forall x [\text{boy}(x) \rightarrow \text{love}(x, (\lambda w' \lambda g' [\llbracket \text{his}_i \text{ mother} \rrbracket^{w'g'}])^{wg[i/x]})]$

= $\forall x [\text{boy}(x) \rightarrow \text{love}(x, \llbracket \text{his}_i \text{ mother} \rrbracket^{wg[i/x]})]$

= $\llbracket [\text{every boy}]_i \text{ loves his}_i \text{ mother} \rrbracket^{wg}$

Basic Idea of Proposal (II)

The semantic way of getting reconstruction effects:

An abstract meaning in one place is contextualized in another place.

Questions

- 1 Why do we need an abstract meaning?
- 2 How does an expression change a context?
- 3 Where or when is an abstract meaning contextualized?

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Basic semantics for the analysis

- Model $M = \langle W, D, F \rangle$, where W is a set of possible worlds;
 D is a set of individuals;
 F is a function from a set of words to a set of meanings.

$$F(\text{love}) = \lambda w' \lambda y \lambda x [x \text{ loves}^{w'} y]$$

$$\text{love}^w = \lambda y \lambda x [x \text{ loves}^w y]$$

- $G = \{g : g \text{ is a function from } N \text{ into } D\}$, where N is a set of natural numbers.

$$[[t_i]]^{wg_1} = g_1(i) = \text{john}, [[\text{he}_j]]^{wg_1} = g_1(j) = \text{bill}$$

$$g_1[i/\text{mary}] = g_2; g_2(i) = \text{mary}$$

Operators and scopes (I)

- The notion of scope applies to operators.
- An operator introduces a new **possible world** or a new **assignment in its scope**.
- A pair of a possible world and an assignment constitutes a **linguistic context**.

Examples of operators:

quantifiers

- $\llbracket [\text{Every } A]_i B \rrbracket^{wg} = \forall x \llbracket [A]^{wg}(x) \rightarrow [B]^{wg[i/x]} \rrbracket$
- $\llbracket [A(n)/\text{Some } A]_i B \rrbracket^{wg} = \exists x \llbracket [A]^{wg}(x) \& [B]^{wg[i/x]} \rrbracket$

Quantifiers introduce new assignments $g[i/x]$.

Operators and scopes (II)

propositional attitude verbs

- $\llbracket A \text{ thinks that } B \rrbracket^{wg} = \forall w' \in \text{Dox}(w, \llbracket A \rrbracket^{wg}) [\llbracket B \rrbracket^{w'g}]$

$\text{Dox}(w, d)$ = a set of possible worlds which are compatible with what d thinks in w

The verb “think” introduces new possible worlds w' .

Proper names and free pronouns (I)

- A proper name or non-anaphoric pronoun can bind something.
- A proper name or non-anaphoric pronoun is not an operator in the sense that they impose conditions on input linguistic contexts, but it is quantifier-raised. (Heim & Kratzer 1998, 245-264)

Suppose α is a proper name or non-anaphoric pronoun.
Then

$[[\alpha_i \beta]]^{wg} = [[\alpha_i]]^{wg}(\lambda x [[\beta]]^{wg[i/x]}),$ where

$[[\alpha_i]]^{wg} = \ll g(i) = d \gg \lambda P[P(g(i))]$ for an individual d

- α_i is semantically bound if $[[\alpha_i]]^{wg[i/d]}$

Heim & Kratzer's definition of semantic binding is still syntactic:

Let α^m be a variable occurrence in a tree γ which is (semantically) bound in γ , and let β^n be a variable binder occurrence in γ .

Then β^n (semantically) binds α^m iff the sister of β^n is the largest subtree of γ in which α^m is (semantically) free.

Proper names and free pronouns (II)

- **procrastination of interpretation of DPs that BCs apply to:**

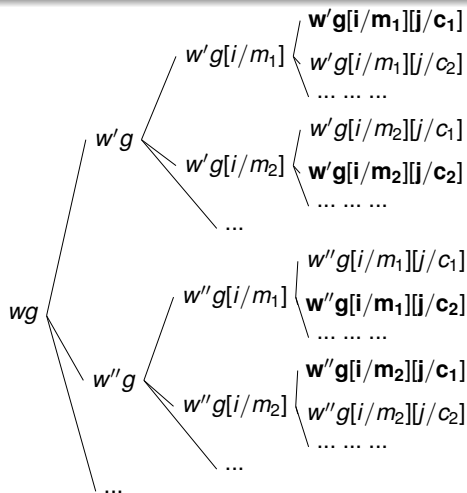
DPs that BCs apply to are interpreted last.

$$\begin{aligned}
 & \llbracket \text{John}_i \text{ } [t_i \text{ washed } \underline{\text{his}}_i \text{ car}] \rrbracket^{wg} \\
 &= \llbracket \text{John}_i \rrbracket^{wg} (\lambda y [\llbracket t_i \text{ washed his}_i \text{ car} \rrbracket^{wg[i/y]}]) \\
 &= \llbracket \text{John}_i \rrbracket^{wg} (\lambda y [\text{washed}^w(y, \llbracket \text{his}_i \text{ car} \rrbracket^{wg[i/y]})]) \\
 &= \llbracket \text{John}_i \rrbracket^{wg} (\lambda y [\text{washed}^w(y, \iota x [\text{car}^w(x) \ \& \ \text{own}^w(\llbracket \underline{\text{his}}_i \rrbracket^{wg[i/y]}, x))])]) \\
 &\Rightarrow \underline{\text{John}}: \text{ free, } \underline{\text{his}}: \text{ bound}
 \end{aligned}$$

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Linguistic contexts



Anaphora

- A car in the embedded context is not available in the matrix clause.

Example 10 (Universal quantifier)

??Every man loves a girl. She is young.

Example 11 (Belief context)

??Amy thinks that John loves a girl. She is young.

Non-contextual meanings (I)

- The set of cars can be captured by what I call **non-contextual meanings**.

$$[[[a \text{ car}]_j^F]]^{wg} = \lambda w' \lambda g' \lambda P \exists y [car^{w'}(y) \ \& \ P(y)]$$

- Expressions that generate alternatives must denote non-contextual meanings, because alternatives must be valid where they are generated and where they are used.
- One alternative of a car in the position of the focus-sensitive operator must be able to denote different cars that every man loves in w' .

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Connectivity of variable binding (I)

- (3) $[\text{What}_j \llbracket [\text{every boy}]_i [t_i \text{ loves } t_j^F] \rrbracket]$ is $[\text{his}_i \text{ mother}]^F$.
 $= [\text{The thing that } \llbracket [\text{every boy}]_i [t_i \text{ loves } t_j^F] \rrbracket \text{ is } [\text{his}_i \text{ mother}]^F]$.

- A gap can be focused.

Example 12 (Focused Gap)

I wrote a letter to Daniel for buying me a book using a fountain pen which_i he **also** bought me $[t_i]^F$

- For a focused gap, we need a **non-contextual assignment** $\mathbf{g}$, in addition to a **contextual assignment** g .
 $\llbracket [t_j^F] \rrbracket^{wg\mathbf{g}} = \mathbf{g}(j)$
- The non-contextual value of the gap is contextualized in the local context because love selects a contextual meaning.
 $\llbracket [\text{love } t_j^F] \rrbracket^{wg\mathbf{g}} = \lambda x [\text{love}^w(x, \mathbf{g}(j)^{wg})]$

Connectivity of variable binding (II)

- The quantifier changes assignments.

$$\llbracket [\text{every boy}]_i [\text{t}_i \text{ loves t}_j^F] \rrbracket^{wgg}$$

$$= \llbracket [\text{every boy}]_i \rrbracket^{wgg} (\lambda y \llbracket [\text{t}_i \text{ loves t}_j^F] \rrbracket^{wg[i/y]g})$$
- The relative clause abstracts over non-contextual meanings

$$\llbracket [\text{that } [\text{every boy}]_i [\text{t}_i \text{ loves t}_j^F] \rrbracket^{wgg}$$

$$= \lambda x \llbracket [\text{every boy}]_i \rrbracket^{wgg} [j/x] (\lambda y \llbracket [\text{t}_i \text{ loves t}_j^F] \rrbracket^{wg[i/y]g[j/x]})$$

$$= \lambda x [\text{every boy}_i \text{ loves } x^{wg[i/y]}]$$
- The subject denotes a definite non-contextual meaning:

$$\llbracket [\text{the thing that } [\text{every boy}]_i \text{ loves t}_j^F] \rrbracket^{wgg} =$$

$$\iota x [\text{every boy}_i \text{ loves } x^{wg[i/y]}] (= \mathbf{A})$$
- The post-copular phrase denotes a non-contextual meaning.

$$\llbracket [\text{his}_i \text{ mother}]^F \rrbracket^{wgg} = \lambda w' \lambda g' \iota y [\text{mother}^{w'}(y, \llbracket [\text{his}_i] \rrbracket^{w'g'g})] (= \mathbf{B})$$

Connectivity of variable binding (III)

- The copula is compatible with a non-contextual meaning.
 $\Rightarrow$ Not contextualized
- The two non-contextual meanings are identified.

$$\mathbf{A} =_{id} \mathbf{B}$$

$$= \iota \mathbf{x} [\text{every}_i \text{ boy loves } \mathbf{x}^{wg[i/y]}] =_{id}$$

$$\lambda w' \lambda g' \iota z [\text{mother}^{w'}(z, [\text{his}_i]^{w'g'g})]$$

$$= [\text{every boy}]_i \text{ loves}$$

$$\lambda w' \lambda g' \iota z [\text{mother}^{w'}(z, [\text{his}_i]^{w'g'g})(w)(g[i/y])]$$

$$= [\text{every boy}]_i \text{ loves } \iota z [\text{mother}^w(z, [\text{his}_i]^{wg[i/y]g})]$$

- The pronoun his is bound by the universal quantifier.

4 kinds of predicates

- This implies that there is a group of words that are compatible with non-contextual meaning.
 - stage-level predicate
 - individual-level predicate
 - kind-level predicate
 - non-contextual meaning level predicate

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BT without reconstruction (I)

- The pronoun he is quantifier-raised.

Example 6

*[[Whose]_k examination of John_i]_j did he_i [t_i fear t_j^[+wh]]?
(Whose exam(ination) of John = the exam of John by whom)

- A wh-phrase, a wh-word, and the trace of a wh-phrase are interpreted as denoting a non-contextual meaning and a wh-word generates a set of alternatives.

Assumption: A wh-question denotes a set of propositions.

$$\llbracket \text{Whose}_k^{[+wh]} \rrbracket^{wgg} = \{ \mathbf{x} : \mathbf{x} \in ALT(\mathbf{g}(k)) \}$$

$$\llbracket [\text{the exam of John}_i \text{ by whom}_k^{[+wh]}]_j \rrbracket^{wgg} = \\ \{ \lambda w' \lambda g' \iota e [exam^{w'}(e, \mathbf{x}^{w'g'}, \llbracket \text{John}_i \rrbracket^{w'g'g})] : \mathbf{x} \in ALT(\mathbf{g}(k)) \} (= \mathbf{A})$$

- The trace of a wh-phrase is contextualized in the local context with fear

$$\llbracket [t_j^{[+wh]}] \rrbracket^{wgg} = \mathbf{g}(j); \llbracket [\text{fear } t_j^F] \rrbracket^{wgg} = \lambda x [fear^w(x, \mathbf{g}(j)^{wgg})]$$

BT without reconstruction (II)

- The TP is assumed to denote a proposition.

$$\begin{aligned} & \llbracket [\text{he}_i \text{ } [t_i \text{ fear } t_j^{[+wh]}]] \rrbracket^{wgg} \\ &= \lambda w' [\llbracket \text{he}_i \rrbracket^{w'gg} (\lambda y [\llbracket t_i \text{ fear } t_j^{[+wh]} \rrbracket^{w'g[i/y]g})] \end{aligned}$$

- M' is interpreted by abstracting over the non-contextual values that the gap can get.

$$\begin{aligned} & \lambda z [\llbracket [M' \text{ did } \text{ } [\text{he}_i \text{ } [t_i \text{ fear } t_j^{[+wh]}]] \rrbracket] \rrbracket^{wgg[j/z]} \\ &= \lambda z \lambda w' [\llbracket \text{he}_i \rrbracket^{w'gg[j/z]} (\lambda y [\llbracket t_i \text{ fear } t_j^{[+wh]} \rrbracket^{w'g[i/y]g[j/z]}))] \\ &= \lambda z \lambda w' [\llbracket \text{he}_i \rrbracket^{w'gg} (\lambda y [\text{fear}^{w'}(y, z^{w'g[i/y]})])] (= \mathbf{B}) \end{aligned}$$

- The meaning of the M' takes each member of the meanings of the preposed wh-phrase.

$$\begin{aligned} \{\mathbf{B}(\mathbf{a}): \mathbf{a} \in \mathbf{A}\} &= \{\lambda w' [\text{fear}^{w'}(g(i), \\ & \quad \iota e [\text{exam}^{w'}(e, \mathbf{x}^{w'g[i/john]}, \llbracket \text{John}_j \rrbracket^{w'g[i/john]g})])], \\ & \quad \mathbf{x} \text{ in ALT}(\mathbf{g}(k))\} \end{aligned}$$

Conclusion

- Expressions that generate alternatives denote non-contextual meanings because their alternatives are used in two different contexts.
- A non-linguistic meaning is defined as an abstraction over world-assignment pairs.
- A linguistic context is defined as a pair of a possible world and an assignment.
- An expression denoting a non-contextual meaning takes a linguistic scope where the non-contextual meaning is contextualized.
- Kaplan's character: abstraction over **non-linguistic** contexts.

Non-contextual meaning: abstraction over **linguistic** contexts

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